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Operational and Embodied Greenhouse-Gas Emissions of Hyperscale
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Findings:

1. Embodied emissions from construction and IT hardware account for 20 to 35 percent of lifecycle greenhouse-gas emissions for hyperscale facilities on moderately clean grids, and up to 55 percent on very low-carbon grids.
2. Median operational emissions intensity across 61 audited facilities was 310 kg $CO_2e$ per MWh of IT load in 2024, with an interquartile range of 120 to 470 kg $CO_2e$ per MWh driven by regional grid mix.
3. Server manufacturing dominates embodied emissions at 58 percent of the embodied total, exceeding concrete and steel combined.
4. Hourly matched carbon-free procurement reduced attributed operational emissions by 38 to 71 percent versus annual certificate matching in the five facilities that disclosed hourly settlement data.
5. The authors note substantial disclosure gaps: only 9 of 61 facilities published third-party-verified embodied-emissions inventories.